We create chemistry

Safety data sheet

1. Identification

Product identifier

Obvius Plus Fungicide Seed Treatment

Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses: crop protection product, fungicide

Details of the supplier of the safety data sheet

Company:
BASF SE
67056 Ludwigshafen
GERMANY
Operating Division Crop Protection

Telephone: +49 621 60-27777
E-mail address: Produktinformation-Pflanzenschutz@basf.com

Emergency telephone number

International emergency number:
Telephone: +49 180 2273-112

2. Hazards Identification

Classification of the substance or mixture

According to UN GHS criteria

Acute Tox. 4 (oral)
Skin Irrit. 3
Muta. 2
Carc. 2
Repr. Additional category for effects on or via lactation.
Repr. 2 (unborn child)
Aquatic Acute 1
Aquatic Chronic 1

For the classifications not written out in full in this section the full text can be found in section 16.

Label elements

Globally Harmonized System (GHS)

Pictogram:

Signal Word:
Warning

Hazard Statement:

H316	Causes mild skin irritation.
H302	Harmful if swallowed.
H362	May cause harm to breast-fed children.
H351	Suspected of causing cancer.
H361	Suspected of damaging the unborn child.
H341	Suspected of causing genetic defects.
H400	Very toxic to aquatic life.
H410	Very toxic to aquatic life with long lasting effects.

Precautionary Statement:

P101	If medical advice is needed, have product container or label at hand.
P102	Keep out of reach of children.
P103	Read carefully and follow all instructions.

Precautionary Statements (Prevention):

P280	Wear protective gloves, protective clothing and eye protection or face protection.
P201	Obtain special instructions before use.
P202	Do not handle until all safety precautions have been read and understood.
P260	Do not breathe dust or mist.
P270	Do not eat, drink or smoke when using this product.
P264	Wash contaminated body parts thoroughly after handling.
P263	Avoid contact during pregnancy and while nursing.

Precautionary Statements (Response):

P308 + P313	IF exposed or concerned: Get medical attention.
P301 + P312	IF SWALLOWED: Call a POISON CENTER or doctor/physician if you feel unwell.
P330	Rinse mouth.
P391	Collect spillage.
P332 + P313	If skin irritation occurs: Get medical attention.

Precautionary Statements (Storage):

P405	Store locked up.

Precautionary Statements (Disposal):

P501	Dispose of contents and container to hazardous or special waste collection point.

Labeling of special preparations (GHS):
May produce an allergic reaction. Contains: metalaxyl (ISO); methyl-N-(2,6-dimethylphenyl)-N-(methoxyacetyl)-DL-alaninate, thiophanate-methyl (ISO); 1,2-di-(3-methoxycarbonyl-2-thioureido) benzene, 1,2-Benzisothiazol-3(2H)-one, mixture of: 5-chloro-2-methyl-2H-isothiazol-3-one and 2-methyl-2H-isothiazol-3-one (3:1)

According to UN GHS criteria

Hazard determining component(s) for labelling: metalaxyl (ISO); methyl-N-(2,6-dimethylphenyl)-N-(methoxyacetyl)-DL-alaninate, thiophanate-methyl (ISO); 1,2-di-(3-methoxycarbonyl-2-thioureido) benzene, 1H-Pyrazole-4-carboxamide, 3-(difluoromethyl)-1-methyl-N-(3',4',5'-trifluoro[1,1'-biphenyl]-2-yl)-; Fluxapyroxad, pyraclostrobin (ISO); methyl N-{2-[1-(4-chlorophenyl)-1H-pyrazol-3-yloxymethyl]phenyl}(N-methoxy)carbamate

Other hazards

According to UN GHS criteria

See section 12 - Results of PBT and vPvB assessment.

If applicable information is provided in this section on other hazards which do not result in classification but which may contribute to the overall hazards of the substance or mixture.

3. Composition/Information on Ingredients

Substances

Not applicable

Mixtures

Chemical nature

crop protection product, fungicide, Flowable concentrate for seed treatment (FS)

Hazardous ingredients (GHS)

According to UN GHS criteria

metalaxyl (ISO); methyl-N-(2,6-dimethylphenyl)-N-(methoxyacetyl)-DL-alaninate

Content (W/W): 14,73 %	Acute Tox. 4 (oral)
CAS Number: 57837-19-1	Skin Sens. 1
	Repr. 2 (unborn child)
	Aquatic Acute 3
	Aquatic Chronic 3
	H302, H317, H361, H402, H412

thiophanate-methyl (ISO); 1,2-di-(3-methoxycarbonyl-2-thioureido) benzene

Content (W/W): 8,93 %	Acute Tox. 4 (Inhalation - dust)
CAS Number: 23564-05-8	Skin Sens. 1
EC-Number: 245-740-7	Muta. 2
	Aquatic Acute 1
	Aquatic Chronic 1
	M-factor acute: 10
	M-factor chronic: 10
	H332, H317, H341, H400, H410

1H-Pyrazole-4-carboxamide, 3-(difluoromethyl)-1-methyl-N-(3',4',5'-trifluoro[1,1'-biphenyl]-2-yl)-; Fluxapyroxad

Content (W/W): 4,46 %	Repr. Add. cat. lact.
CAS Number: 907204-31-3	Aquatic Acute 1
	Aquatic Chronic 1
	M-factor acute: 1
	M-factor chronic: 1
	H362, H400, H410

pyraclostrobin (ISO); methyl N-{2-[1-(4-chlorophenyl)-1H-pyrazol-3-yloxymethyl]phenyl}(N-methoxy)carbamate

Content (W/W): 3,56 %	Acute Tox. 3 (Inhalation - mist)
CAS Number: 175013-18-0	Acute Tox. 4 (oral)
INDEX-Number: 613-272-00-6	Skin Irrit. 2
	Repr. 2 (unborn child)
	STOT SE 3 (irr. to respiratory syst.)
	STOT RE (Liver, Nasal cavity, Gastrointestinal tract) 2
	Aquatic Acute 1
	Aquatic Chronic 1
	M-factor acute: 100
	M-factor chronic: 100
	H315, H331, H302, H335, H361, H373, H400, H410

1,2-benzisothiazol-3(2H)-one

Content (W/W): < 0,05 % CAS Number: 2634-33-5 EC-Number: 220-120-9 INDEX-Number: 613-088-00-6	Acute Tox. 2 (Inhalation - dust) Acute Tox. 4 (oral) Skin Irrit. 2 Eye Dam. 1 Skin Sens. 1A Aquatic Acute 1 Aquatic Chronic 1 M-factor acute: 1 M-factor chronic: 1 H318, H315, H330, H302, H317, H400, H410 <u>Specific concentration limit:</u> Skin Sens. 1A: >= 0,036 %

Bronopol

Content (W/W): < 0,05 % CAS Number: 52-51-7 EC-Number: 200-143-0 INDEX-Number: 603-085-00-8	Acute Tox. 3 (Inhalation - dust) Acute Tox. 3 (oral) Acute Tox. 4 (dermal) Skin Irrit. 2 Eye Dam. 1 STOT SE 3 (irr. to respiratory syst.) Aquatic Acute 1 Aquatic Chronic 1 M-factor acute: 100 M-factor chronic: 10 H318, H315, H312, H335, H301 + H331, H400, H410

mixture of: 5-chloro-2-methyl-2H-isothiazol-3-one and 2-methyl-2H-isothiazol-3-one (3:1)

Content (W/W): < 0,0015 % CAS Number: 55965-84-9 INDEX-Number: 613-167-00-5	Acute Tox. 3 (oral) Acute Tox. 2 (Inhalation - mist) Acute Tox. 2 (dermal) Skin Corr. 1C Eye Dam. 1 Skin Sens. 1A Aquatic Acute 1 Aquatic Chronic 1 M-factor acute: 100 M-factor chronic: 100 H301, H317, H314, H310 + H330, H400, H410 EUH071 <u>Specific concentration limit:</u> Skin Sens. 1A: >= 0,0015 % Eye Dam./Irrit. 1: >= 0,6 % Eye Dam./Irrit. 2: 0,06 - < 0,6 % Skin Corr./Irrit. 1C: >= 0,6 % Skin Corr./Irrit. 2: 0,06 - < 0,6 %

Glycerol

Content (W/W): < 10 %
CAS Number: 56-81-5
EC-Number: 200-289-5

For the classifications not written out in full in this section the full text can be found in section 16.

4. First-Aid Measures

Description of first aid measures

Remove contaminated clothing.

If inhaled:
Keep patient calm, remove to fresh air, seek medical attention.

On skin contact:
Wash thoroughly with soap and water

On contact with eyes:
Wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

On ingestion:
Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

Most important symptoms and effects, both acute and delayed

Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11., (Further) symptoms and / or effects are not known so far

Hazards: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11. (Further) symptoms and / or effects are not known so far

Indication of any immediate medical attention and special treatment needed

Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

5. Fire-Fighting Measures

Extinguishing media

Suitable extinguishing media:
water spray, dry powder, foam, carbon dioxide

Special hazards arising from the substance or mixture

Carbon monoxide, Carbon dioxide, hydrogen chloride, Hydrogen fluoride, Hydrogen bromide, nitrogen oxides, sulfur oxides, halogenated compounds

The substances/groups of substances mentioned can be released in case of fire.

Advice for fire-fighters

Special protective equipment:
Wear self-contained breathing apparatus and chemical-protective clothing.

Further information:
Keep containers cool by spraying with water if exposed to fire. In case of fire and/or explosion do not breathe fumes. Collect contaminated extinguishing water separately, do not allow to reach sewage or effluent systems. Dispose of fire debris and contaminated extinguishing water in accordance with official regulations.

6. Accidental Release Measures

Personal precautions, protective equipment and emergency procedures

Do not breathe vapour/spray. Use personal protective clothing. Avoid contact with the skin, eyes and clothing.

Environmental precautions

Do not discharge into the subsoil/soil. Do not discharge into drains/surface waters/groundwater.

Methods and material for containment and cleaning up

For small amounts: Pick up with suitable absorbent material (e.g. sand, sawdust, general-purpose binder, kieselguhr).
For large amounts: Dike spillage. Pump off product.
Dispose of absorbed material in accordance with regulations. Collect waste in suitable containers, which can be labeled and sealed. Clean contaminated floors and objects thoroughly with water and detergents, observing environmental regulations. Wear suitable protective equipment.

7. Handling and Storage

Precautions for safe handling

No special measures necessary if stored and handled correctly. Ensure thorough ventilation of stores and work areas. When using do not eat, drink or smoke. Hands and/or face should be washed before breaks and at the end of the shift.

Protection against fire and explosion:
No special precautions necessary. The substance/product is non-combustible. Product is not explosive.

Conditions for safe storage, including any incompatibilities

Segregate from foods and animal feeds.
Further information on storage conditions: Keep away from heat. Protect from direct sunlight.

Storage stability:
Storage duration: 24 Months

Specific end use(s)

For the relevant identified use(s) listed in Section 1 the advice mentioned in this section 7 is to be observed.

8. Exposure Controls/Personal Protection

Control parameters

Components with occupational exposure limits

56-81-5: Glycerol
23564-05-8: Thiophanate-methyl
TWA value 0,86 mg/m3 (BASF recomm. occupational exposure limit)
57837-19-1: Methyl N-(2,6-dimethylphenyl)-N-(methoxyacetyl)-DL-alaninate
175013-18-0: Carbamic acid, [2-[[[1-(4-chlorophenyl)-1H-pyrazol-3-yl]oxy]methyl]phenyl]methoxy-, methyl ester
TWA value 0,13 mg/m3 (BASF recomm. occupational exposure limit)
907204-31-3: Fluxapyroxad
TWA value 0,5 mg/m3 (BASF recomm. occupational exposure limit)

Exposure controls

Personal protective equipment

Respiratory protection:
Suitable respiratory protection for higher concentrations or long-term effect: Combination filter for gases/vapours of organic, inorganic, acid inorganic and alkaline compounds (e.g. EN 14387 Type ABEK).

Hand protection:
Suitable chemical resistant safety gloves (EN ISO 374-1) also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN ISO 374-1): E.g. nitrile rubber (0.4 mm), chloroprene rubber (0.5 mm), butyl rubber (0.7 mm) etc.

Eye protection:
Safety glasses with side-shields (frame goggles) (e.g. EN 166)

Body protection:
Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures
The statements on personal protective equipment in the instructions for use apply when handling crop-protection agents in final-consumer packing. Wearing of closed work clothing is recommended. Store work clothing separately. Keep away from food, drink and animal feeding stuffs.

9. Physical and Chemical Properties

9.1. Information on basic physical and chemical properties

State of matter: liquid
Form: liquid
Colour: white
Odour: odourless
Odour threshold:
not applicable, odour not perceivable
Melting point:
The product has not been tested.
Boiling point: approx. 100 °C
Information applies to the solvent.
Flammability: not applicable
Lower explosion limit:
As a result of our experience with this product and our knowledge of its composition we do not expect any hazard as long as the product is used appropriately and in accordance with the intended use.
Upper explosion limit:
As a result of our experience with this product and our knowledge of its composition we do not expect any hazard as long as the product is used appropriately and in accordance with the intended use.
Flash point:
No flash point - Measurement made up to the boiling point.
Auto-ignition temperature: 455 °C
Thermal decomposition: No decomposition if stored and handled as prescribed/indicated.
pH value: approx. 6,5 - 8,5
(1 %(m), 20 °C)
Viscosity, dynamic: approx. 89 mPa.s
(20 °C, 100 1/s)
Solubility in water: dispersible
Partitioning coefficient n-octanol/water (log Kow):
not applicable
Information on: metalaxyl (ISO); methyl-N-(2,6-dimethylphenyl)-N-(methoxyacetyl)-DL-alaninate
Partitioning coefficient n-octanol/water (log Kow): 1,65
Information on: thiophanate-methyl (ISO); 1,2-di-(3-methoxycarbonyl-2-thioureido) benzene
Partitioning coefficient n-octanol/water (log Kow): approx. (OECD Guideline 107)
1,44
(25 °C)

Vapour pressure: approx. 23 hPa
(20 °C)
Information applies to the solvent.
Density: approx. 1,12 g/cm3
(20 °C)
Relative vapour density (air):
not applicable

9.2. Other information

Information with regard to physical hazard classes

Explosives
Explosion hazard: not explosive
Oxidizing properties
Fire promoting properties: not fire-propagating

Other safety characteristics

Other Information: If necessary, information on other physical and chemical parameters is indicated in this section.
Evaporation rate:
not applicable

10. Stability and Reactivity

Reactivity

No hazardous reactions if stored and handled as prescribed/indicated.

Chemical stability

The product is stable if stored and handled as prescribed/indicated.

Possibility of hazardous reactions

No hazardous reactions if stored and handled as prescribed/indicated.

Conditions to avoid

See SDS section 7 - Handling and storage.

Incompatible materials

Substances to avoid:
strong acids, strong bases, strong oxidizing agents

Hazardous decomposition products

Hazardous decomposition products:
No hazardous decomposition products if stored and handled as prescribed/indicated.

11. Toxicological Information

Information on toxicological effects

Acute toxicity

Assessment of acute toxicity:

Of moderate toxicity after single ingestion. Virtually nontoxic by inhalation. Virtually nontoxic after a single skin contact.

Experimental/calculated data:
LD50 rat (oral): > 500 - < 2.000 mg/kg

LC50 rat (by inhalation): > 2,251 mg/l
No mortality was observed. An aerosol was tested.

LD50 rat (dermal): > 5.000 mg/kg
No mortality was observed.

Irritation

Assessment of irritating effects:
Skin contact causes slight irritation. Not irritating to the eyes.

Experimental/calculated data:
Skin corrosion/irritation rabbit: Slightly irritating.

Serious eye damage/irritation rabbit: non-irritant

Respiratory/Skin sensitization

Assessment of sensitization:
No sensitizing effect.

Experimental/calculated data:
Buehler test guinea pig: Non-sensitizing.

Germ cell mutagenicity

Assessment of mutagenicity:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: thiophanate-methyl (ISO); 1,2-di-(3-methoxycarbonyl-2-thioureido) benzene
Assessment of mutagenicity:
Mutagenic properties can not be excluded on the basis of experimental data.

Information on: Bronopol
Assessment of mutagenicity:
The substance was not mutagenic in bacteria. The substance was mutagenic in various cell culture test systems; however, these results could not be confirmed in tests with mammals.

Carcinogenicity

Assessment of carcinogenicity:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: 1H-Pyrazole-4-carboxamide, 3-(difluoromethyl)-1-methyl-N-(3',4',5'-trifluoro[1,1'-biphenyl]-2-yl)-; Fluxapyroxad

Assessment of carcinogenicity:
Indication of possible carcinogenic effect in animal tests. The effect is caused by an animal specific mechanism that has no human counter part.

Information on: thiophanate-methyl (ISO); 1,2-di-(3-methoxycarbonyl-2-thioureido) benzene
Assessment of carcinogenicity:
The results of various animal studies gave no indication of a carcinogenic effect.

Reproductive toxicity

Assessment of reproduction toxicity:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: 1H-Pyrazole-4-carboxamide, 3-(difluoromethyl)-1-methyl-N-(3',4',5'-trifluoro[1,1'-biphenyl]-2-yl)-; Fluxapyroxad
Assessment of reproduction toxicity:
The results of animal studies gave no indication of a fertility impairing effect. May cause harm to children via breast-feeding.

Developmental toxicity

Assessment of teratogenicity:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: pyraclostrobin (ISO); methyl N-{2-[1-(4-chlorophenyl)-1H-pyrazol-3-yloxymethyl]phenyl}(N-methoxy)carbamate
Assessment of teratogenicity:
Indications of possible developmental toxicity/teratogenicity were seen in animal studies.

Information on: metalaxyl (ISO); methyl-N-(2,6-dimethylphenyl)-N-(methoxyacetyl)-DL-alaninate
Assessment of teratogenicity:
Indications of possible developmental toxicity/teratogenicity were seen in animal studies.

Specific target organ toxicity (single exposure)

Assessment of STOT single:
Based on the available information there is no specific target organ toxicity to be expected after a single exposure.

Remarks: The product has not been tested. The statement has been derived from the properties of the individual components.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: 1H-Pyrazole-4-carboxamide, 3-(difluoromethyl)-1-methyl-N-(3',4',5'-trifluoro[1,1'-biphenyl]-2-yl)-; Fluxapyroxad

Assessment of repeated dose toxicity:
Adaptive effects were observed after repeated exposure in animal studies.

Information on: pyraclostrobin (ISO); methyl N-{2-[1-(4-chlorophenyl)-1H-pyrazol-3-yloxymethyl]phenyl}(N-methoxy)carbamate
Assessment of repeated dose toxicity:
Repeated exposure may affect certain organs. Target organs: Liver, gastrointestinal tract and nasal cavity

Information on: Bronopol
Assessment of repeated dose toxicity:
After repeated exposure the prominent effect is local irritation.

Information on: mixture of: 5-chloro-2-methyl-2H-isothiazol-3-one and 2-methyl-2H-isothiazol-3-one (3:1)
Assessment of repeated dose toxicity:
After repeated exposure the prominent effect is local irritation. Based on available data, the classification criteria are not met.

Aspiration hazard

not applicable

Other relevant toxicity information

Misuse can be harmful to health.

12. Ecological Information

Toxicity

Assessment of aquatic toxicity:
Very toxic to aquatic life with long lasting effects.
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: pyraclostrobin (ISO); methyl N-{2-[1-(4-chlorophenyl)-1H-pyrazol-3-yloxymethyl]phenyl}(N-methoxy)carbamate
Toxicity to fish:
LC50 (96 h) 0,00616 mg/l, Oncorhynchus mykiss (EPA 72-1, Flow through.)

Information on: 1H-Pyrazole-4-carboxamide, 3-(difluoromethyl)-1-methyl-N-(3',4',5'-trifluoro[1,1'-biphenyl]-2-yl)-; Fluxapyroxad
Toxicity to fish:
LC50 (96 h) 0,29 mg/l, Cyprinus carpio (Fish test acute, semistatic)

LC50 (96 h) 0,546 mg/l, Oncorhynchus mykiss (OECD Guideline 203, static)

LC50 (96 h) 1,15 mg/l, Lepomis macrochirus (OECD Guideline 203, static)

LC50 (96 h) 0,466 mg/l, Pimephales promelas (OECD Guideline 203, static)

Information on: metalaxyl (ISO); methyl-N-(2,6-dimethylphenyl)-N-(methoxyacetyl)-DL-alaninate
Toxicity to fish:
LC50 (96 h) 27 mg/l, Lepomis macrochirus

Information on: thiophanate-methyl (ISO); 1,2-di-(3-methoxycarbonyl-2-thioureido) benzene
Toxicity to fish:
LC50 (96 h) 1,07 mg/l, Oncorhynchus mykiss (static)

Information on: pyraclostrobin (ISO); methyl N-{2-[1-(4-chlorophenyl)-1H-pyrazol-3-yloxymethyl]phenyl}(N-methoxy)carbamate
Aquatic invertebrates:
EC50 (48 h) 0,0157 mg/l, Daphnia magna (OECD Guideline 202, part 1, static)

EC50 (96 h) 0,00416 mg/l, Americamysis bahia

Information on: 1H-Pyrazole-4-carboxamide, 3-(difluoromethyl)-1-methyl-N-(3',4',5'-trifluoro[1,1'-biphenyl]-2-yl)-; Fluxapyroxad
Aquatic invertebrates:
EC50 (48 h) 6,78 mg/l, Daphnia magna (OECD Guideline 202, part 1, static)

Information on: metalaxyl (ISO); methyl-N-(2,6-dimethylphenyl)-N-(methoxyacetyl)-DL-alaninate
Aquatic invertebrates:
LC50 12,5 mg/l, Daphnia magna

Information on: thiophanate-methyl (ISO); 1,2-di-(3-methoxycarbonyl-2-thioureido) benzene
Aquatic invertebrates:
EC50 (48 h) 5,4 mg/l, Daphnia magna (Flow through.)

Information on: pyraclostrobin (ISO); methyl N-{2-[1-(4-chlorophenyl)-1H-pyrazol-3-yloxymethyl]phenyl}(N-methoxy)carbamate
Aquatic plants:
EC10 (7 d) 0,82 mg/l (growth rate), Lemna gibba

EC50 (7 d) > 1,007 mg/l (growth rate), Lemna gibba

EC50 (72 h) 0,011 mg/l (growth rate), Navicula pelliculosa

Information on: 1H-Pyrazole-4-carboxamide, 3-(difluoromethyl)-1-methyl-N-(3',4',5'-trifluoro[1,1'-biphenyl]-2-yl)-; Fluxapyroxad
Aquatic plants:
EC50 (72 h) 0,70 mg/l (growth rate), Pseudokirchneriella subcapitata (OECD Guideline 201)

EC50 (96 h) 0,66 mg/l (growth rate), Pseudokirchneriella subcapitata

EC10 (72 h) 0,31 mg/l (growth rate), Pseudokirchneriella subcapitata

EC10 (96 h) 0,36 mg/l (growth rate), Pseudokirchneriella subcapitata

Information on: thiophanate-methyl (ISO); 1,2-di-(3-methoxycarbonyl-2-thioureido) benzene
Aquatic plants:
EC50 (72 h) > 25,4 mg/l (growth rate), Chlorella vulgaris (OECD Guideline 201)

No observed effect concentration (72 h) 4,38 mg/l, Pseudokirchneriella subcapitata

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Information on: pyraclostrobin (ISO); methyl N-{2-[1-(4-chlorophenyl)-1H-pyrazol-3-yloxymethyl]phenyl}(N-methoxy)carbamate
Chronic toxicity to fish:
No observed effect concentration (98 d) approx. 0,00235 mg/l, Oncorhynchus mykiss (OECD Guideline 210, Flow through.)

Information on: 1H-Pyrazole-4-carboxamide, 3-(difluoromethyl)-1-methyl-N-(3',4',5'-trifluoro[1,1'-biphenyl]-2-yl)-; Fluxapyroxad
Chronic toxicity to fish:
No observed effect concentration (33 d) 0,0359 mg/l, Pimephales promelas (OECD Guideline 210, Flow through.)

Information on: thiophanate-methyl (ISO); 1,2-di-(3-methoxycarbonyl-2-thioureido) benzene
Chronic toxicity to fish:
No observed effect concentration (28 d) 0,32 mg/l, Oncorhynchus mykiss
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Information on: pyraclostrobin (ISO); methyl N-{2-[1-(4-chlorophenyl)-1H-pyrazol-3-yloxymethyl]phenyl}(N-methoxy)carbamate
Chronic toxicity to aquatic invertebrates:
No observed effect concentration (21 d) 0,004 mg/l, Daphnia magna (OECD Guideline 202, part 2, semistatic)
The details of the toxic effect relate to the nominal concentration.

No observed effect concentration (31 d) 0,000365 mg/l, Mysidopsis bahia

Information on: 1H-Pyrazole-4-carboxamide, 3-(difluoromethyl)-1-methyl-N-(3',4',5'-trifluoro[1,1'-biphenyl]-2-yl)-; Fluxapyroxad
Chronic toxicity to aquatic invertebrates:
No observed effect concentration (21 d) 0,5 mg/l, Daphnia magna (OECD Guideline 211, semistatic)

Information on: thiophanate-methyl (ISO); 1,2-di-(3-methoxycarbonyl-2-thioureido) benzene
Chronic toxicity to aquatic invertebrates:
EC50 (21 d) 0,18 mg/l, Daphnia magna

Persistence and degradability

Assessment biodegradation and elimination (H2O):
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: pyraclostrobin (ISO); methyl N-{2-[1-(4-chlorophenyl)-1H-pyrazol-3-yloxymethyl]phenyl}(N-methoxy)carbamate
Assessment biodegradation and elimination (H2O):
Not readily biodegradable (by OECD criteria).

Information on: 1H-Pyrazole-4-carboxamide, 3-(difluoromethyl)-1-methyl-N-(3',4',5'-trifluoro[1,1'-biphenyl]-2-yl)-; Fluxapyroxad
Assessment biodegradation and elimination (H2O):
Not readily biodegradable (by OECD criteria).

Information on: metalaxyl (ISO); methyl-N-(2,6-dimethylphenyl)-N-(methoxyacetyl)-DL-alaninate
Assessment biodegradation and elimination (H2O):
Not readily biodegradable (by OECD criteria).

Information on: thiophanate-methyl (ISO); 1,2-di-(3-methoxycarbonyl-2-thioureido) benzene
Assessment biodegradation and elimination (H2O):
Not readily biodegradable (by OECD criteria).

Bioaccumulative potential

Assessment bioaccumulation potential:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: metalaxyl (ISO); methyl-N-(2,6-dimethylphenyl)-N-(methoxyacetyl)-DL-alaninate
Assessment bioaccumulation potential:
Because of the n-octanol/water distribution coefficient (log Pow) accumulation in organisms is not to be expected.

Information on: thiophanate-methyl (ISO); 1,2-di-(3-methoxycarbonyl-2-thioureido) benzene
Assessment bioaccumulation potential:
Because of the n-octanol/water distribution coefficient (log Pow) accumulation in organisms is not to be expected.

Information on: pyraclostrobin (ISO); methyl N-{2-[1-(4-chlorophenyl)-1H-pyrazol-3-yloxymethyl]phenyl}(N-methoxy)carbamate
Bioaccumulation potential:
Bioconcentration factor: 379 - 507, Oncorhynchus mykiss (OECD Guideline 305)
Accumulation in organisms is not to be expected.

Information on: 1H-Pyrazole-4-carboxamide, 3-(difluoromethyl)-1-methyl-N-(3',4',5'-trifluoro[1,1'-biphenyl]-2-yl)-; Fluxapyroxad
Bioaccumulation potential:
Bioconcentration factor: 36 - 37 (28 d), Lepomis macrochirus (OECD Guideline 305)
Does not accumulate in organisms.

Mobility in soil

Assessment transport between environmental compartments:
Adsorption in soil: The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: pyraclostrobin (ISO); methyl N-{2-[1-(4-chlorophenyl)-1H-pyrazol-3-yloxymethyl]phenyl}(N-methoxy)carbamate
Assessment transport between environmental compartments:
Adsorption in soil: Following exposure to soil, adsorption to solid soil particles is probable, therefore contamination of groundwater is not expected.

Information on: 1H-Pyrazole-4-carboxamide, 3-(difluoromethyl)-1-methyl-N-(3',4',5'-trifluoro[1,1'-biphenyl]-2-yl)-; Fluxapyroxad

Assessment transport between environmental compartments:
Adsorption in soil: Following exposure to soil, adsorption to solid soil particles is probable, therefore contamination of groundwater is not expected.

Information on: thiophanate-methyl (ISO); 1,2-di-(3-methoxycarbonyl-2-thioureido) benzene
Assessment transport between environmental compartments:
Adsorption in soil: Following exposure to soil, adsorption to solid soil particles is probable, therefore contamination of groundwater is not expected.

Results of PBT and vPvB assessment

The product does not contain a substance fulfilling the PBT (persistent/bioaccumulative/toxic) criteria or the vPvB (very persistent/very bioaccumulative) criteria.

Other adverse effects

The product does not contain substances that are listed in the Montreal Protocol on substances that deplete the ozone layer.

Additional information

Other ecotoxicological advice:
Do not discharge product into the environment without control.

13. Disposal Considerations

Waste treatment methods

Must be sent to a suitable incineration plant, observing local regulations.

Contaminated packaging:
Contaminated packaging should be emptied as far as possible and disposed of in the same manner as the substance/product.

14. Transport Information

Land transport

ADR

UN number or ID number:	UN3082
UN proper shipping name:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (THIOPHANATE-METHYL, PYRACLOSTROBIN)
Transport hazard class(es):	9, EHSM
Packing group:	III

Environmental hazards:	yes
Special precautions for user:	None known

RID

UN number or ID number:	UN3082
UN proper shipping name:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (THIOPHANATE-METHYL, PYRACLOSTROBIN)
Transport hazard class(es):	9, EHSM
Packing group:	III
Environmental hazards:	yes
Special precautions for user:	None known

Inland waterway transport

ADN

UN number or ID number:	UN3082
UN proper shipping name:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (THIOPHANATE-METHYL, PYRACLOSTROBIN)
Transport hazard class(es):	9, EHSM
Packing group:	III
Environmental hazards:	yes
Special precautions for user:	None known

Transport in inland waterway vessel
Not evaluated

Sea transport

IMDG

UN number or ID number:	UN 3082
UN proper shipping name:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (THIOPHANATE-METHYL, PYRACLOSTROBIN)
Transport hazard class(es):	9, EHSM
Packing group:	III
Environmental hazards:	yes Marine pollutant: YES
Special precautions for user:	EmS: F-A; S-F

Air transport

IATA/ICAO

UN number or ID number:	UN 3082
UN proper shipping name:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (THIOPHANATE-METHYL, PYRACLOSTROBIN)
Transport hazard class(es):	9, EHSM
Packing group:	III
Environmental hazards:	yes
Special precautions for user:	None known

Maritime transport in bulk according to IMO instruments

Maritime transport in bulk is not intended.

Further information

Product may be shipped as non-hazardous in suitable packages containing a net quantity of 5 L or less under the provisions of various regulatory agencies: ADR, RID, ADN: Special Provision 375; IMDG: 2.10.2.7; IATA: A197; TDG: Special Provision 99(2); 49CFR: §171.4 (c) (2) and also the Special Provision 375 in Appendix B which is regulated in China "Regulations Concerning Road Transportation of Dangerous Goods Part 3: Index of dangerous goods name and transportation requirements" (JT/T 617.3)

15. Regulatory Information

Safety, health and environmental regulations/legislation specific for the substance or mixture

To avoid risks to man and the environment, comply with the instructions for use.

16. Other Information

Full text of classifications, hazard symbols and hazard statements, if mentioned in section 2 or 3:

Acute Tox.	Acute toxicity
Skin Irrit.	Skin irritation
Muta.	Germ cell mutagenicity
Carc.	Carcinogenicity
Repr.	Reproductive toxicity
Aquatic Acute	Hazardous to the aquatic environment - acute
Aquatic Chronic	Hazardous to the aquatic environment - chronic
Skin Sens.	Skin sensitization
STOT SE	Specific target organ toxicity — single exposure
STOT RE	Specific target organ toxicity — repeated exposure
Eye Dam.	Serious eye damage
Skin Corr.	Skin corrosion
Eye Dam./Irrit.	Serious eye damage/eye irritation
Skin Corr./Irrit.	Skin corrosion/irritation
H302	Harmful if swallowed.
H317	May cause an allergic skin reaction.
H361	Suspected of damaging the unborn child.
H402	Harmful to aquatic life.

H412	Harmful to aquatic life with long lasting effects.
H332	Harmful if inhaled.
H341	Suspected of causing genetic defects.
H400	Very toxic to aquatic life.
H410	Very toxic to aquatic life with long lasting effects.
H362	May cause harm to breast-fed children.
H315	Causes skin irritation.
H331	Toxic if inhaled.
H335	May cause respiratory irritation.
H373	May cause damage to organs (Liver, Nasal cavity, Gastrointestinal tract) through prolonged or repeated exposure.
H318	Causes serious eye damage.
H330	Fatal if inhaled.
H312	Harmful in contact with skin.
H301 + H331	Toxic if swallowed or if inhaled.
H301	Toxic if swallowed.
H314	Causes severe skin burns and eye damage.
H310 + H330	Fatal in contact with skin or if inhaled.
EUH071	Corrosive to the respiratory tract.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.